

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

INCREMENTAL MODERNIZATION OF A LEGACY
UNIVERSITY TIMETABLE WEB APPLICATION
DIPLOMA THESIS

2027

BC. KRISTÍNA RYBÁROVÁ

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UNIVERSITY TIMETABLE WEB APPLICATION
DIPLOMA THESIS

Study Programme: Applied Computer Science
Field of Study: Computer Science
Department: Department of Applied Informatics
Supervisor: Mgr. Marek Šuppa

Bratislava, 2027
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Univerzita Komenského v Bratislave
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Študijný program: aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor: informatika
Typ záverečnej práce: diplomová
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Incremental Modernization of a Legacy University Timetable Web Application
Postupná modernizácia webovej aplikácie pre rozvrh hodín na univerzite

Anotácia:

Cieľ:

Literatúra:

Vedúci: Mgr. Marek Šuppa
Katedra: FMFI.KAI - Katedra aplikovanej informatiky
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Dátum zadania: 12.12.2025

Dátum schválenia: 13.12.2025

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vedúci práce



THESIS ASSIGNMENT

Name and Surname:	Bc. Kristína Rybárová
Study programme:	Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study:	Computer Science
Type of Thesis:	Diploma Thesis
Language of Thesis:	English
Secondary language:	Slovak
Title:	Incremental Modernization of a Legacy University Timetable Web Application
Annotation:	Candle is a production academic web application for timetable browsing and personal schedule construction operated by a student team at FMFI UK, currently implemented as a legacy PHP/Symfony 1.x system with multiple integration points (e.g., authentication and calendar export). This thesis aims to perform an engineering case study of porting Candle to a modern stack in 2025 while preserving externally observable behavior, using an incremental displacement strategy (Strangler Fig) rather than a big-bang rewrite. The work will emphasize regression-safety through a migration-oriented test strategy (characterization/approval tests plus a balanced automated test portfolio), guided by contemporary testing practice and evaluated not only via coverage but also via mutation testing to estimate fault-detection capability as the system is progressively replatformed. The study design will follow established case study methodology to triangulate technical artifacts and outcomes (repository mining, test metrics, pipeline telemetry, and developer feedback) into transferable guidance for modernizing small-team academic production systems.
Aim:	The goals of the thesis include (but are not limited to): - Characterizing Candle's current architecture, dependencies, and modernization risk profile based on the public repository and production constraints. - Designing and implementing a regression-safety strategy centered on characterization/approval tests and a balanced automated test portfolio aligned with the practical test pyramid. - Executing an incremental migration plan (facade + progressive replacement) and documenting decision trade-offs versus rewrite-oriented alternatives. - Building a modern CI pipeline with repeatable environments and automated quality gates suitable for ongoing student-team maintenance. - Evaluating migration outcomes using mutation testing and other engineering metrics, and synthesizing generalizable lessons using a case study protocol.
Literature:	Fowler, Martin. "Strangler Fig Application." Fowler, Martin. "The Practical Test Pyramid." Sánchez, A. B., et al. "Mutation Testing in Practice" Feathers, Michael. Working Effectively with Legacy Code Mohottige, T. I., et al. "Reengineering software systems into microservices: State-of-the-art and future directions"



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Declaration: I hereby declare that I have independently written this entire bachelor thesis entitled “Incremental Modernization of a Legacy University Timetable Web Application”, including all its figures and appendices, using the literature listed in the attached bibliography.

In preparation of the thesis I have also used the following artificial intelligence tools: [TOOLS] for the following purposes: [PURPOSES]. I have used artificial intelligence tools in accordance with relevant legal regulations, academic rights and freedoms, ethical and moral principles while simultaneously maintaining academic integrity. I am aware that I am fully responsible for the accuracy of the final text.

Acknowledgments: Tu môžete poďakovať školiteľovi, prípadne ďalším osobám, ktoré vám s prácou nejako pomohli, poradili, poskytli dáta a podobne.

Abstrakt

Slovenský abstrakt v rozsahu 100-500 slov, jeden odstavec. Abstrakt stručne sumarizuje výsledky práce. Mal by byť pochopiteľný pre bežného informatika. Nemal by teda využívať skratky, termíny alebo označenie zavedené v práci, okrem tých, ktoré sú všeobecne známe.

Kľúčové slová: jedno, druhé, tretie (prípadne štvrté, piate)

Abstract

Abstract in the English language (translation of the abstract in the Slovak language).

Keywords:

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Introduction

See file `uvod.tex` for information about recommended contents of the Introduction chapter.

Chapter 1

Legacy System Modernization

First paper[1]

Second paper[2]

Chapter 2

Case Study Design and Migration Strategy

Chapter 3

Architecture and CI/CD Pipeline Design

Chapter 4

Implementation Results and Engineering Metrics

Conclusion

See file `zaver.tex` for information about recommended contents of the Conclusion chapter.

Bibliography

- [1] Thakshila Imiya Mohottige, Artem Polyvyanyy, Colin Fidge, Rajkumar Buyya, and Alistair Barros. Reengineering software systems into microservices: State-of-the-art and future directions. 183:107732.
- [2] Ana B. Sánchez, José A. Parejo, Sergio Segura, Amador Durán, and Mike Papadakis. Mutation testing in practice: Insights from open-source software developers. 50(5):1130–1143.